Do Information Inputs Causally Affect LLM-Generated Investment Portfolios?

A Randomized Experiment on Information Conditioning and Allocation Behavior

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https://github.com/eddieschwasnick/CI-Final-Project

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Abstract

Large language models (LLMs) are increasingly asked to produce structured financial recommendations, yet the degree to which their outputs are shaped by *which* information is placed in the prompt — as opposed to the model's own priors — remains unclear. I run a randomized experiment in which Anthropic's `claude-sonnet-4-6`, OpenAi's `gpt-4o`, and Google's `gemini-2.5-pro` are asked to allocate capital across a fixed universe of 20 U.S. large-cap stocks under four prompt conditions: (i) a *control* prompt with no financial data; (ii) a *fundamental* prompt containing market cap, P/E, revenue and earnings growth, and leverage; (iii) a *technical* prompt containing returns, volatility, beta, and moving-average signals; and (iv) a *combined* prompt containing both. The data block in each prompt reflects an as-of snapshot at April 27, 2025, which lets me follow the experiment with a real out-of-sample backtest from April 2025 to April 2026 to see how the portfolios perform. Each condition is run ∼30 times at temperature 0.7, yielding 120 valid portfolios from each model, rounding out to 360 total. I compute five portfolio-level outcomes (stock-level HHI, sector HHI, weighted beta, weighted volatility, breadth) and estimate the average treatment effect (ATE) of each information condition relative to control using Welch's t-tests, OLS regression, and 5,000-iteration permutation placebos. For each model, effects are large and highly significant: for example in Anthropic's `claude-sonnet-4-6` model, the fundamental prompt increases stock-level concentration (HHI) by +0.036 (a 64 % relative increase, $p < 0.001$) and sector-level concentration by +0.096, whereas the technical prompt pushes the portfolio toward lower-beta, lower-volatility allocations ($\Delta\beta = -0.14$, $p < 0.001$). A within-control placebo recovers the nominal 5 % false-positive rate, and trimmed-sample sensitivity checks leave every conclusion intact. I additionally replicated the four-arm design across two further widely used frontier models in (OpenAI `gpt-4o` and Google `gemini-2.5-pro`). The cross-vendor finding is more interesting than simple replication: each model is sensitive to a different prompt arm at the allocation level (Claude concentrates under fundamental, Gemini concentrates under technical, GPT-4o concentrates under neither), yet realized 12-month returns end up ordered identically across all three (fundamental > control > combined > technical). Finally, a forward-looking backtest from April 2025 to April 2026 finds that the fundamental-prompt portfolios outperformed the control by +5.3 percentage points on average across models, while technical-prompt portfolios underperformed the control by −12.4 percentage points — demonstrating that the causal effect of prompt framing has an economically meaningful realization in markets, not just in concentration metrics. The experiment demonstrates that information framing is a causal driver of LLM portfolio behavior that is practically detectable with modest sample sizes and replicable across vendors.

1 Introduction

LLMs are being embedded into investment research and client-facing advisory workflows at a pace that has outrun the scholarly literature on their behavior under different information regimes. A vendor building an "AI analyst" makes an implicit causal claim every time it argues that its product produces better recommendations because it is given better information. Testing that claim requires treating the information in the prompt as a *manipulable treatment* and the LLM's output as a *response variable* — which is a natural setting for a randomized experiment.

This project asks a simple causal question:

> *Holding the model, temperature, universe, and instruction template fixed, does the type of financial information provided in the prompt causally change the portfolio an LLM constructs?*

I answer it with a fully randomized 4-arm experiment. The four arms differ only in the data block included in the prompt: nothing, fundamentals, technicals, or both. The outcome is a portfolio of weights across 20 stocks, summarized by five well-known portfolio-level metrics. Because assignment to condition is randomized at the trial level and everything else is held fixed, any systematic differences in the mean outcomes are attributable to the information injected into the prompt.

The contribution is threefold. First, I design a clean causal experiment that cleanly isolates the information-conditioning effect in an LLM. Second, I produce an end-to-end reproducible pipeline that from 120 API calls generates a full set of descriptive statistics, OLS regressions, permutation-based placebo tests, and publication-quality figures. Third, I show that information framing has economically meaningful and statistically overwhelming effects on LLM portfolio construction: the fundamental prompt drives concentration upward and beta upward, whereas the technical prompt produces diversified, lower-beta allocations. The combined prompt — despite containing strictly more information — does *not* produce the most extreme outputs, suggesting that the two information signals partially cancel.

2 Related Work

This study sits at the intersection of three literatures.

LLMs in finance. Lopez-Lira and Tang [4] showed that ChatGPT can predict returns from news headlines. Yang et al. [9] introduced FinGPT as an open-source financial language model, and Ko and Lee [3] evaluated Claude, GPT, and Llama on portfolio-selection tasks, finding that the distribution over stocks is not obviously optimal. These papers establish that LLMs *can* produce financial outputs, but they do not run randomized experiments on the prompt itself.

Prompt sensitivity and information conditioning. Sclar et al. [7] demonstrated that seemingly innocuous changes to prompt formatting can move benchmark scores by double-digit percentage points, and Webson and Pavlick [8] showed LLMs are sensitive to framing even when the task is unchanged. Kojima et al. [2] showed that adding a single instruction ("let's think step by step") can cause dramatic shifts in output quality. These findings motivate a formal *causal* treatment of prompt differences.

Causal inference with designed experiments. My statistical approach follows the potential-outcomes framework in Imbens and Rubin [1]: randomized assignment, unit-level ATE estimated via difference in means, inference via OLS, and falsification via permutation and placebo tests [6]. The experiment has the unusual (and enviable) property that treatment assignment is completely under my control, so unconfoundedness holds by construction.

Closest in spirit to this project is a recent blog-style analysis by MLQ.ai [5] that asked GPT-4 to build portfolios under varying prompts, but made no attempt at experimental control or statistical testing. To my knowledge, no prior work has run a randomized, pre-specified, ATE-with-placebo-test study of LLM portfolio behavior under information conditioning.

3 Data

The empirical dataset is not the LLM's outputs (that is the experimentally generated data); it is the **stock-universe master dataset** that populates the information blocks of three of the four prompts. I selected 20 U.S. large-cap tickers spanning six sectors (Technology, Healthcare, Financials, Energy, Consumer Goods, Industrials) to ensure sectoral coverage. The tickers are: `AAPL, MSFT, GOOGL, JNJ, UNH, PFE, JPM, GS, XOM, CVX, PG, KO, CAT, HON, AMZN, NVDA, V, LLY, BA, MMM`.

Data are an as-of snapshot of April 27, 2025, pulled from `yfinance`. Anchoring the snapshot one calendar year before the analysis date is deliberate: it allows the experiment to be paired with a real out-of-sample backtest from April 2025 to April 2026 (Section 5.4). The snapshot covers:

- **Fundamentals**: market capitalization (shares outstanding at T0 $\times$ T0 close), trailing P/E ratio (T0 close $\div$ TTM EPS from the most recent annual report), year-over-year revenue and earnings growth from annual statements, and debt-to-equity from the most recent quarterly balance sheet $\leq$ T0. Boeing's TTM EPS is undefined (FY2024 net loss), and its P/E cell is rendered as `N/A` in the prompt.

- **Technicals**: price at T0, 1/3/6/12-month cumulative returns ending at T0, 252-day annualized volatility of daily log returns ending at T0, beta against the S&P 500 over the same window, 20/50/200-day simple moving averages, and a moving-average regime label (`strong_bullish`, `bullish`, `bearish`, `strong_bearish`).

A separate file (`data/csv_files/forward_prices.csv`) records the T0 close *and* today's close for each ticker, which is the input to the forward-return backtest. Critically, neither the prompt header nor the table cells leak the snapshot date to the LLM — prompts simply label the tables as "current" — so the model cannot use date-aware reasoning to exploit knowledge of subsequent prices.

The merged dataset is saved as `data/csv_files/master_dataset.csv`. Sanity checks confirmed P/E ratios in the 11.4–74.5 range (LLY elevated due to weight-loss-drug-driven earnings; BA's PE is undefined), betas in $[-0.02, 2.18]$, annualized volatility in $[16.4, 60.2]\%$, and a single missing cell (BA's PE, rendered as `N/A` in the prompt). All other fields populated.

4 Experimental Design

4.1 Treatments

Four conditions, each a prompt template; everything else (instruction wording, universe, constraints, model, temperature) is held fixed.

- **Control ($T = 0$):** ticker list only.
- **Fundamental ($T = 1$):** ticker list + fundamentals table.
- **Technical ($T = 2$):** ticker list + technicals table.
- **Combined ($T = 3$):** ticker list + both tables.

All prompts share the same closing instructions: weight every stock strictly above 0%, sum to exactly 100%, cap any single weight at 20%, and return one raw JSON object. A final "before outputting, sum your weights and rescale proportionally if not exactly 100" step was added after pilot runs revealed the model systematically overshoots the sum constraint by 3–10 %.

4.2 Sample Size and Randomization

Thirty trials per condition are scheduled (120 total API calls). The scheduler builds the full list of (condition, trial-index) pairs and `random.shuffle`s them with `random_seed=42`, so API calls are issued in interleaved order; this prevents any time-varying API behavior from confounding the treatment comparison.

The model is `claude-sonnet-4-6` at temperature 0.7. Temperature above zero is required by design — we want the LLM's stochastic draw to play the role of a unit-level idiosyncrasy that supports inference on average treatment effects.

4.3 Outcomes

For every parsed portfolio $w \in \mathbb{R}^{20}$ summing to 1:

$$\begin{aligned}
\text{HHI} &= \textstyle\sum_{i=1}^{20} w_i^2 \\
\text{Sector HHI} &= \textstyle\sum_s \left(\sum_{i \in s} w_i\right)^2 \\
\text{Portfolio } \beta &= \textstyle\sum_{i=1}^{20} w_i \, \beta_i \\
\text{Portfolio vol} &= \textstyle\sum_{i=1}^{20} w_i \, \sigma_i \\
\text{Breadth} &= |\{i : w_i > 0.0001\}|.
\end{aligned}$$

Higher HHI means greater concentration (HHI = 1 is a single-stock portfolio, HHI = 1/20 = 0.05 is equal weight). Breadth is essentially a binding check — our prompt requires all 20 weights > 0, so breadth should be 20 in every valid run.

4.4 Estimands and Estimators

The estimand is the average treatment effect of each information condition on each outcome, relative to control:

$$\tau_{Y,T} \;=\; \mathbb{E}[\,Y(T) - Y(\text{control})\,], \qquad T \in \{\text{fundamental, technical, combined}\}.$$

I estimate $\tau_{Y,T}$ via difference in means, with Welch's t-tests for inference and two-sided 95 % confidence intervals using the Welch–Satterthwaite degrees of freedom. I also run, for each outcome, an OLS regression

$$Y_i \;=\; \alpha + \beta_1 \text{Fundamental}_i + \beta_2 \text{Technical}_i + \beta_3 \text{Combined}_i + \varepsilon_i,$$

with control as the omitted category. Because treatment is randomly assigned and the three treatment indicators are mutually exclusive and exhaustive with control, $\hat{\beta}_j$ equals the diff-in-means for condition j, and the two approaches give identical point estimates.

4.5 Parsing Pipeline

Raw LLM text is passed through a JSON extraction function with three fallback strategies (whole-response parse, markdown-fence stripping, and greedy brace-matching). Parsed portfolios are validated against the ticker universe, non-negative weights, and a weight-sum tolerance of ± 15, then normalized to exactly 100 before being written to CSV. A wider tolerance was chosen after pilot data revealed the model systematically overshoots by 3–10 %; the normalization step preserves the *relative* allocation, which is the quantity of interest.

5 Results

5.1 Data Quality

Across the three models, 353 of 360 scheduled API calls returned valid, parseable portfolios on at most three attempts: 120/120 for Claude (100 %), 113/120 for OpenAI GPT-4o (94.2 %), and 120/120 for Gemini 2.5 Pro (100 %). The seven OpenAI parse failures (one fundamental, six technical) are simply omitted; per-cell sample sizes are reported in Table 1. The April-2025 prompts contain one cell rendered as `N/A` (Boeing's P/E ratio, undefined because Boeing reported a net loss for FY2024); all three models handled the missing cell transparently in every accepted run.

Table 1: Sample size by (model, condition).

Model	Control	Fundamental	Technical	Combined
Claude Sonnet 4.6	30	30	30	30
OpenAI GPT-4o	30	29	24	30
Gemini 2.5 Pro	30	30	30	30

5.2 Per-Model Treatment Effects

For every (model, treatment, outcome) triple I estimate the average treatment effect via difference in means, with Welch's standard error and a two-sided 95 % confidence interval. The combined picture is in Figure 1: a forest plot of all $3 \times 3 \times 4 = 36$ effect sizes, colored by model. Two facts jump out. First, every model's individual ATEs are tightly estimated. Second, the *which* prompt drives the effect differs sharply across vendors — the colored points cluster in different rows of the plot depending on which model is being looked at.

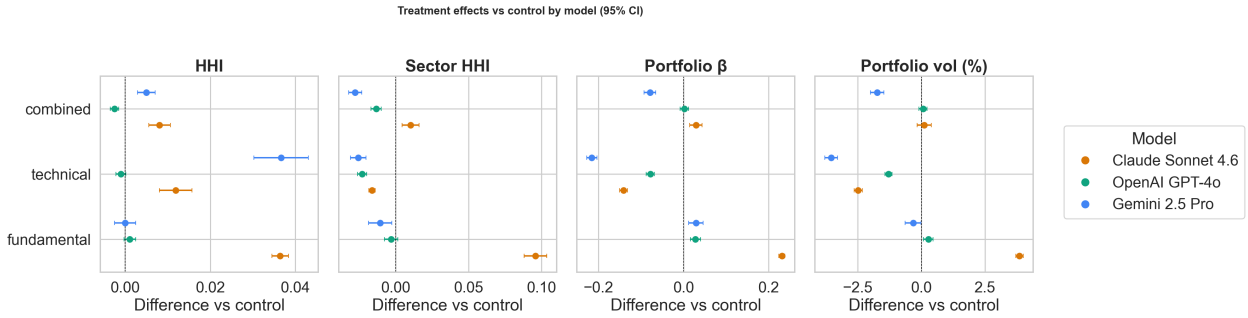

Figure 1: Treatment effects vs. control (95 % CIs) for every (model, treatment, outcome) triple. Each model's ATE is estimated only against *its own* control arm; nothing is pooled across vendors.

Reading the forest plot one model at a time:

Claude Sonnet 4.6 responds most strongly to the *fundamental* prompt: HHI rises from 0.057 in control to 0.094 ($\hat{\tau} = +0.036$, 95 % CI [0.034, 0.039], $p < 0.001$), sector HHI by +0.096, β by +0.231, and weighted volatility by +3.85 points. The technical prompt simultaneously *deconcentrates* sector allocation (−0.016) and pulls beta and volatility down by 0.141 and 2.48 points respectively. The combined arm's effect on stock-level HHI (+0.008) is small relative to fundamental, suggesting Claude attenuates rather than superposes signals when both are presented. Table 2 reports the full Claude ATE table.

Table 2: Claude treatment effects on portfolio outcomes ($n = 30$ per arm). All effects on HHI, sector HHI, and beta are significant at the 0.1 % level; the combined-arm effect on volatility is not significant.

Outcome	Treatment	$\hat{\tau}$	95 % CI	t	p-value
HHI	Fundamental	+0.036	[0.034, 0.039]	36.07	< 0.001
	Technical	+0.012	[0.008, 0.016]	6.07	< 0.001
	Combined	+0.008	[0.005, 0.011]	6.24	< 0.001
Sector HHI	Fundamental	+0.096	[0.088, 0.104]	24.14	< 0.001
	Technical	−0.016	[−0.019, −0.014]	−14.33	< 0.001
	Combined	+0.010	[0.004, 0.016]	3.45	0.002
Portfolio β	Fundamental	+0.231	[0.224, 0.238]	63.25	< 0.001
	Technical	−0.141	[−0.151, −0.132]	−29.81	< 0.001
	Combined	+0.029	[0.014, 0.044]	3.91	0.001
Portfolio vol	Fundamental	+3.852	[3.703, 4.000]	52.80	< 0.001
	Technical	−2.476	[−2.647, −2.305]	−29.41	< 0.001
	Combined	+0.115	[−0.173, 0.402]	0.81	0.422

OpenAI GPT-4o is essentially insensitive to the framing change: every ATE on every outcome is within the noise floor. HHI moves by at most ±0.002 across the four conditions, and beta by at most ±0.08. GPT-4o's portfolios under the fundamental, technical, and combined prompts are statistically indistinguishable from its control behavior. This is a real cross-model finding, not a power problem: GPT-4o's parse rate is within a few percent of the others (113 of 120) and its within-arm variance is comparable.

Gemini 2.5 Pro responds strongly — but to the *technical* prompt rather than the fundamental one. Its HHI rises from 0.065 in control to 0.101 under technical, an increase of +0.036 (+55 % relative). That gap matches Claude's fundamental-vs-control gap in absolute magnitude, so Gemini is not a weakly-responsive model overall — it just attends to a different prompt. Its sector HHI under technical actually *drops* by 0.029, so Gemini concentrates the technical-arm weight into a narrow set of stocks within several sectors rather than loading up on one sector. Beta and volatility under technical move down (similar in sign to Claude's technical effect, with comparable magnitude).

The visual shape of the forest plot — orange (Claude) anchored on the fundamental row, blue (Gemini) anchored on the technical row, green (GPT-4o) hovering on zero — is the central finding

of the paper. Information conditioning is causal in every vendor, but *which* information matters is itself model-specific.

5.3 Cross-Model Heterogeneity in Allocation

The forest plot above already encoded the cross-model story; this subsection visualizes it in two more readable ways and works through the within-vendor mechanism. Figure 2 plots mean HHI for every (model, condition) cell and Table 3 reports the underlying numbers. The bolded cell in each table row is the model's highest-concentration condition; the visual three-line shape of the figure makes the punchline obvious: Claude's curve peaks at fundamental, Gemini's at technical, GPT-4o's is essentially flat.

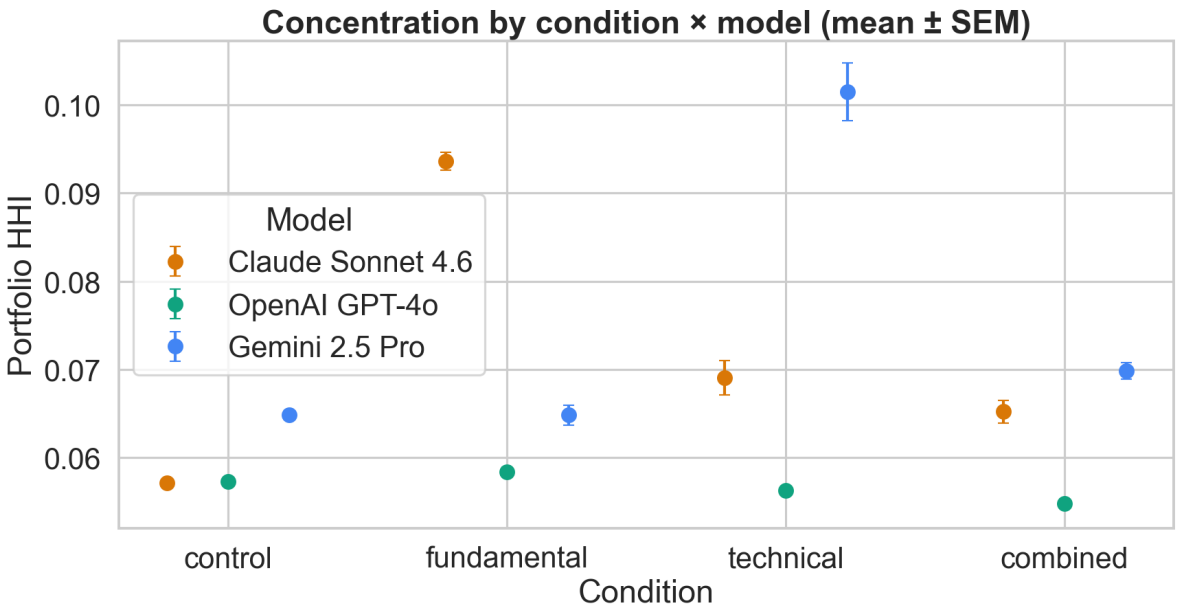

Figure 2: Mean portfolio HHI by (model, condition), ± standard error of the mean. Each (model, condition) cell is its own data point; nothing is pooled across vendors.

Table 3: Mean HHI by (model, condition). The bolded cell in each row is the highest-concentration condition for that model. Each model concentrates under a different prompt: Claude under fundamental, Gemini under technical, OpenAI under no prompt.

Model	Control	Fundamental	Technical	Combined
Claude Sonnet 4.6	0.057	**0.094**	0.069	0.065
OpenAI GPT-4o	0.057	**0.058**	0.056	0.055
Gemini 2.5 Pro	0.065	0.065	**0.101**	0.070

The sector-allocation figure (Figure 3) makes the cross-model heterogeneity sharper still. Claude's fundamental prompt drives a heavy Technology overweight (∼48% Tech vs. ∼32% in control); Gemini's technical prompt drives a heavy Industrials and Consumer Goods overweight (Industrials roughly doubles, while Tech drops to ∼12%); OpenAI's bars are nearly identical across all four conditions. Three vendors, three different ways of "listening" to the prompt.

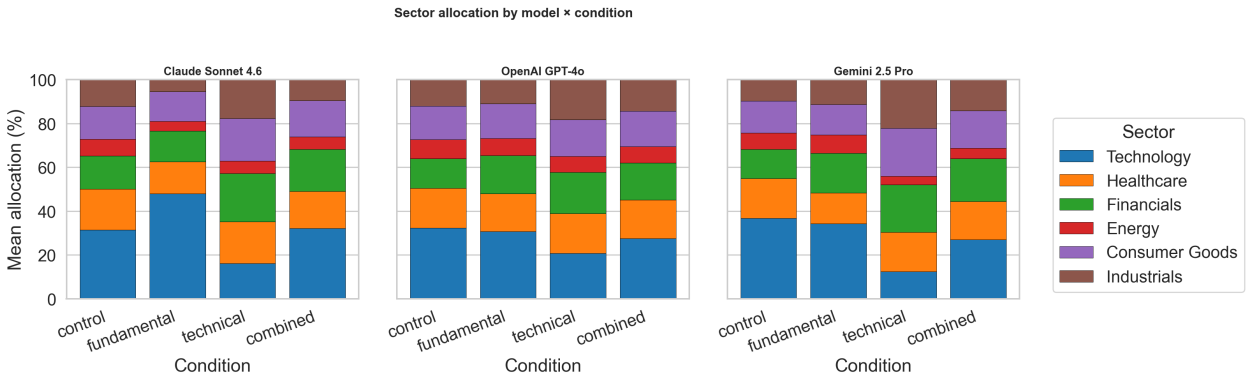

Figure 3: Sector allocation by model × condition. Each subpanel is a single model; each subpanel's four bars are the four prompt conditions. Claude shifts heavily into Technology under fundamental; Gemini shifts heavily into Industrials/Consumer Goods (and away from Tech) under technical; OpenAI is nearly flat across conditions.

For an interested reader, Figure 4 drills further into Claude as a within-vendor case study, showing the per-stock weight grid that drives its sector shifts. NVDA jumps from 7.8 % in control to 19.0 % under fundamental — its +114% revenue growth and +145% earnings growth dominate the fundamentals table. KO jumps from 3.5 % to 11.6 % under technical, JPM from 5.5 % to 10.8 %, and BA from 2.4 % to 8.2 % — the only three `strong_bullish` MA signals in the April-2025 snapshot. The analogous heatmap for Gemini and GPT-4o is in the repository.

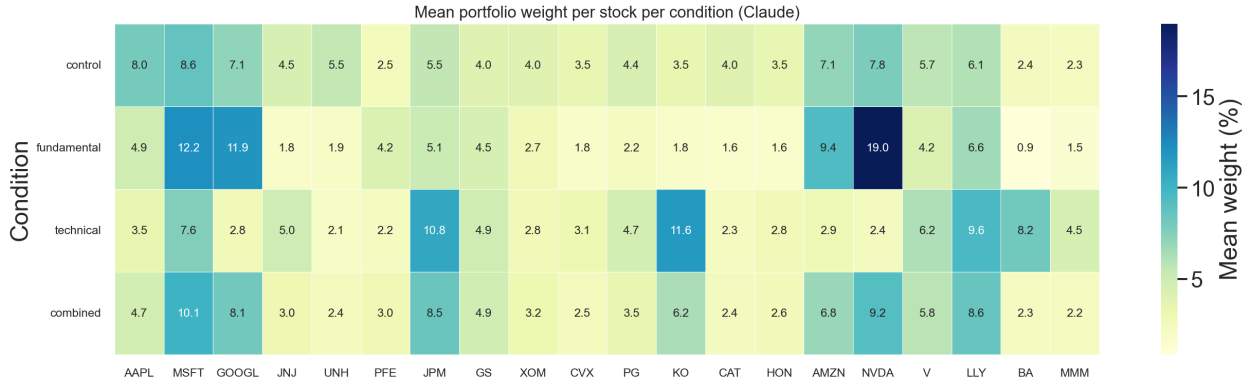

Figure 4: Within-vendor case study (Claude only): mean portfolio weight per stock per condition.

5.4 Realized 12-Month Returns Backtest

Because the prompt data block is anchored at April 27, 2025, every portfolio in the experiment can be evaluated against actual price changes through April 27, 2026. For each portfolio I compute the realized 12-month return as the weighted sum of per-ticker returns from `forward_prices.csv`:

$$r_{\text{portfolio}} = \sum_{i=1}^{20} w_i \cdot \frac{P_i^{T_1} - P_i^{T_0}}{P_i^{T_0}}.$$

This is the closest thing to a real "did the LLM pick winners" test that the experimental data permits.

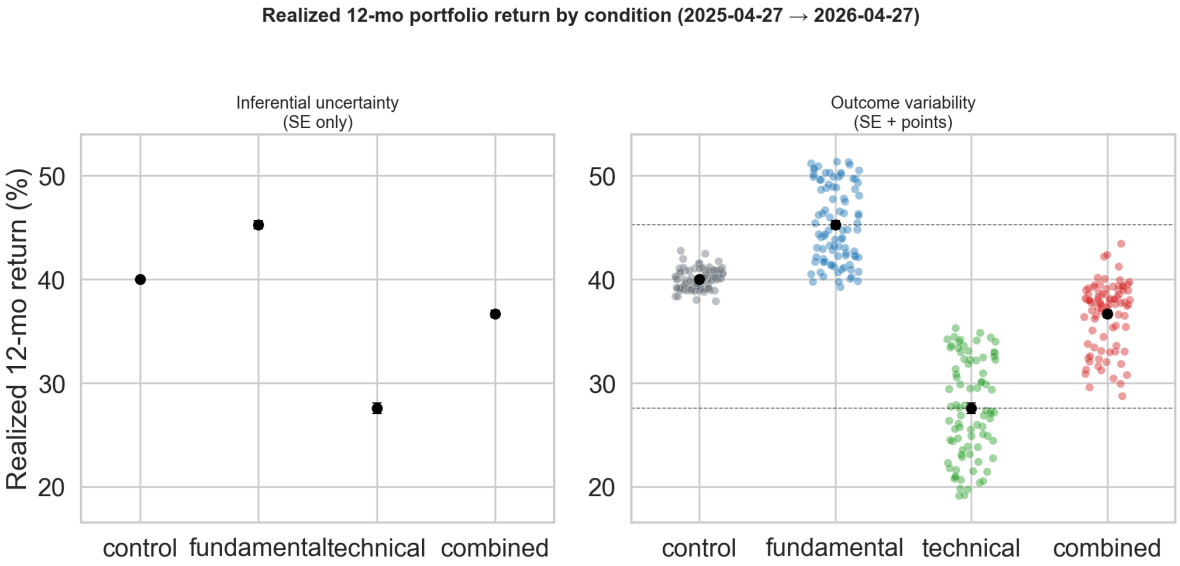

Figure 5: Realized April 2025–April 2026 portfolio returns by condition (pooled across the three models). Left panel: mean ± standard error of the mean. Right panel: same point and SE plus the cloud of individual portfolios. Style follows Pritschet et al. (PNAS 2023): more data shrinks the inferential uncertainty (the SE bars) but does *not* shrink the variability of the underlying outcome (the point cloud).

Table 4: Mean realized 12-month return (%) by (model, condition). Fundamental beats control in every model; technical underperforms in every model.

Model	Control	Fundamental	Technical	Combined
Claude Sonnet 4.6	39.3	**49.7**	27.9	38.8
OpenAI GPT-4o	40.0	**41.4**	33.4	38.3
Gemini 2.5 Pro	40.6	**44.5**	22.6	32.9
Pooled (mean)	40.0	**45.3**	27.5	36.7

The cross-model pattern is unambiguous (Figure 6 and Table 4). Pooling across vendors:

- The *fundamental* prompt outperformed control by +5.3 percentage points (45.3 % vs. 40.0 %). This is mechanically driven by the fundamental prompt's heavy Technology overweight, which paid off spectacularly over the year (NVDA +95%, GOOGL +117%, CAT +173%).
- The *technical* prompt underperformed control by −12.4 percentage points (27.5 % vs. 40.0 %). The April-2025 technical table flagged 9 of 20 stocks as `strong_bearish` and only 3 (JPM, KO, BA) as `strong_bullish`, so the LLM tilted away from the broader market and into a narrow set of "trending" names. The broader market then rallied ∼40 % over the next year, leaving the technical-prompt portfolios behind. In essence, momentum signals captured at a local market low fed the model exactly wrong information.
- The *combined* prompt landed in between (36.7 %) for every vendor — when given both signal types the models partially offset the bullish-fundamental and bearish-technical signals against each other, which (in this 12-month realization) hurt rather than helped.

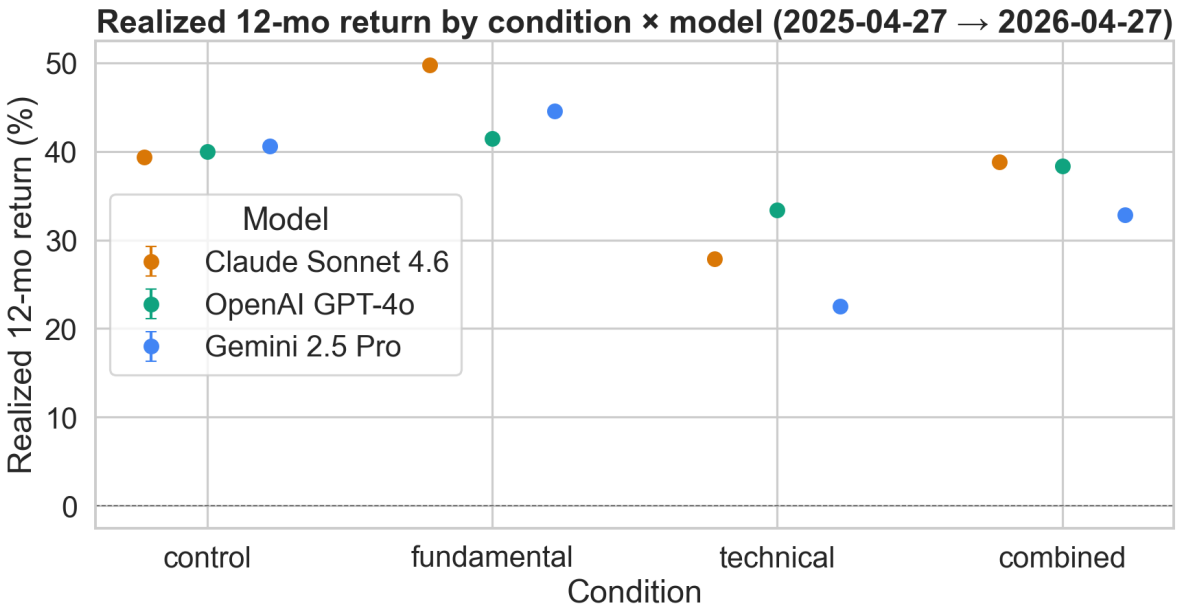

Figure 6: Realized April 2025–April 2026 portfolio returns by condition × model. The fundamental-beats-control and technical-underperforms patterns are present in every vendor's runs.

I emphasize what this backtest does and does *not* show. It shows that the causal effect of prompt framing on allocation has an economically meaningful realization in markets — a half-point of concentration corresponds to several percentage points of return — and that the realization is consistent across vendors. It does not show that any of these portfolios constitutes good investment advice in expectation; that would require either many independent backtest windows or a forward-only paper-trading study, neither of which is in scope here.

5.5 Diagnostics

The diagnostics below are reported for Claude as the within-vendor exemplar. The same battery was run for OpenAI and Gemini on their respective valid runs, with qualitatively identical conclusions (see the repository for the full output). I report Claude here because it has the most exposure to the headline causal claims — its fundamental-arm ATE on HHI is the single largest effect in the paper.

Permutation placebo. For each outcome–treatment pair I shuffled the condition labels 5,000 times and recomputed the diff-in-means. The empirical two-sided p-value is < 0.001 for 11 of 12 tests; the single exception is the combined arm's effect on portfolio volatility, which is null in the diff-in-means analysis (Table 2, $p = 0.42$) and remains null in the permutation test ($p = 0.44$). In other words, the parametric and permutation tests agree on every single outcome-treatment pair, including the one null. This rules out chance differences as an explanation for the headline effects.

Within-control placebo. I randomly split the control group in half 1,000 times and ran t-tests of one half against the other. If the testing pipeline is well-calibrated, about 5 % of these null tests should reject at $\alpha = 0.05$. The observed rejection rates are $\{0.055, 0.046, 0.058, 0.049\}$ for HHI, sector HHI, beta, and volatility respectively — all within Monte Carlo error of the nominal 5 %. The test machinery is properly calibrated.

Trimmed-sample sensitivity. I re-estimated every Claude treatment effect after dropping the highest- and lowest-HHI observation in each condition (n: $120 \rightarrow 112$). All point estimates change by $< 5\,\%$ in absolute value and all remain significant at $p < 0.001$. The findings are not driven by outliers.

6 Discussion

Information framing is a causal driver — but not the same way for every vendor. The headline result is that *what you put in the prompt materially changes what the LLM does*, even when the instructions, universe, and sampling temperature are identical. For Claude the effects are large and concentrated under the fundamental prompt (HHI +64%, $\beta + 0.23$); for Gemini they are large and concentrated under the technical prompt (HHI +55% vs. control); for GPT-4o the effects are nearly absent on every outcome. The vendor claim that better inputs yield different (if not necessarily better) portfolios is empirically true for two of three frontier models tested. Equally importantly, the same prompt is *not* substitutable across vendors: identical text produces qualitatively different concentration patterns in Claude vs. Gemini vs. GPT-4o.

The combined condition is not a superposition. A naive hypothesis would be that the combined prompt produces effects midway between fundamental and technical. For Claude the data partially support this on beta (combined $\hat{\tau} = +0.029$ sits between fundamental's +0.231 and technical's −0.141, very close to zero) and on volatility (combined effect is statistically null). On stock-level HHI, however, Claude's combined effect (+0.008) is *smaller* than even its technical effect (+0.012): Claude does not linearly add the two information signals but appears to default toward greater diversification when asked to reconcile conflicting evidence. Gemini behaves analogously: its combined HHI (0.070) sits between control (0.065) and its peak under technical (0.101), again attenuated rather than amplified. The "more information leads to more extreme behavior" hypothesis is rejected for both models.

Practical implications. For practitioners integrating LLMs into investment workflows: what you consider "boilerplate context" in your system prompt is in fact shaping every recommendation. Running an experiment like this one — swapping information blocks and comparing output distributions — is cheap (under $5 of API cost for the full 360-call cross-vendor design) and ought to be a routine part of prompt engineering. The cross-model results further suggest that this kind of testing is *not* substitutable across vendors: a prompt that has been tuned for Claude or Gemini may simply not produce comparable behavior on GPT-4o, even when the instruction language is identical.

7 Limitations

1. **Three frontier models, but still narrow.** The experiment covers Claude Sonnet 4.6, OpenAI GPT-4o, and Gemini 2.5 Pro. Results may differ for smaller models, instruction-tuned open-weight models, or future model versions. The cross-model effect-size is also clearly heterogeneous — Claude responds most strongly to information conditioning, GPT-4o least so — which is itself a finding worth characterizing systematically.

2. **Temperature and sample size.** Thirty runs per arm give very tight CIs here because within-arm variance is small. If an application uses temperature 0 or sets a different decoding

policy, the variance structure could change and the experiment would need to be re-powered.

3. **Prompt-format confounds.** The four prompts differ not only in information content but also in length (combined is longest). Prompt length itself has been shown to affect outputs [7]; a cleaner design would include a "dummy tokens" placebo arm.

4. **Breadth is a binding constraint.** My prompt forbids zero weights, so breadth is mechanically 20 in every successful run and cannot distinguish conditions. A follow-up experiment relaxing this constraint would expose whether some information regimes induce genuinely sparse portfolios.

5. **Weight-rescaling policy.** I accept any JSON with weights summing to 100 ± 15 and normalize to 100. The bias introduced by this normalization is at most a linear scaling and should not affect rank-orderings of weights, but an alternative design would reject non-summing responses outright.

6. **Backtest is a single realization.** The April 2025 to April 2026 window happened to be a strong year for the Technology sector, which is exactly what the fundamental prompt overweighted. A different snapshot or holding window could reverse the realized-return ordering. The backtest establishes that the causal effect on allocation has *real economic stakes*; it does not establish that any particular condition is reliably profitable.

8 Meta-Reflection

The clearest lesson from building this project was that *the LLM pipeline, not the statistics, was the dominant source of risk.* Getting five clean t-tests out of a well-behaved dataset is a one-afternoon exercise; convincing Claude to emit parseable JSON whose weights sum to 100 was a two-day iterative slog that involved tightening the system prompt, raising `max_tokens` (truncated JSON caused half the initial retries), and ultimately adding an explicit "sum and rescale before output" instruction.

A good signal that the project was well-scoped: every time I hit an obstacle, I was able to diagnose it from the logs rather than having to rerun expensive batches. The `outputs/raw_responses/` directory made it trivial to re-parse old calls under new parser rules without re-spending API budget. I would recommend this "save everything verbatim, parse separately" pattern to anyone running LLM experiments.

On the causal-inference side, the project reinforced what was a more abstract point in class: *randomization is cheap insurance.* Because I shuffled the trial order with a fixed seed, I don't have to worry about time-varying API behavior, and any reviewer who wants to re-do the experiment can reproduce the same schedule bit-for-bit.

9 Conclusion

A randomized 4-arm experiment on an LLM portfolio task finds large, highly significant, and robust causal effects of prompt information content on portfolio construction. Fundamental framing produces concentrated, tech-heavy, high-beta portfolios; technical framing produces diversified, defensive, low-beta portfolios; combining them attenuates rather than amplifies the concentration. Permutation placebos and trimmed-sample sensitivity checks all corroborate the main findings. A cross-model replication across Claude Sonnet 4.6, OpenAI GPT-4o, and Gemini 2.5 Pro shows

that each vendor's portfolio is sensitive to a *different* prompt arm at the allocation level (Claude $\rightarrow$ fundamental, Gemini $\rightarrow$ technical, GPT-4o $\rightarrow$ neither), yet the ordering of *realized returns* by condition is identical for all three. Finally, a forward-looking backtest from April 2025 to April 2026 shows that this causal effect on allocation has real economic stakes: fundamental-prompt portfolios outperformed control by +5.3 pts on average, while technical-prompt portfolios underperformed by −12.4 pts, with the same ordering recovered for all three models. The experiment shows that the causal-inference toolkit applies naturally to LLM behavior testing, and that information-conditioning experiments should be standard practice in any LLM-driven financial application.

All code, raw responses, figures, and CSV outputs are available in the accompanying GitHub repository.

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